

Letters to the Editor.

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Breeding Experiments on the Inheritance of Acquired Characters.

[At the request of the Editor of NATURE, and of Dr. Kammerer, I have translated this letter from the original German into English. Dr. Kammerer has also sent me the typewritten script of a reply to Mr. Cunningham, but in an accompanying letter he tells me that he considers it superfluous to publish this now, as he is quite satisfied with the reply which I made to Mr. Cunningham in my letter to NATURE published on June 23. E. W. MACBRIDE.]

To begin with, may I remark that I have not seen Mr. Bateson's first criticisms of my work (NATURE, July 3, 1919, p. 344) which he cites in the course of his recent letter. Whilst I was in England, my colleagues informed me of the contents of Mr. Bateson's letter of 1919, but I had unfortunately no opportunity of referring to this letter myself. I regret this all the more, since if what I had been informed of its contents was actually in the letter, it would not have been possible for me to enter into any discussion of the subject with Mr. Bateson himself. I must therefore for the present confine myself to his most recent letter (NATURE, June 2, p. 738) and the remarks which he made on the discussion which followed my lecture to the Linnean Society,¹ in which he expressly apologised to me in case I had considered his previous attacks too rude.

It is indeed remarkable that Mr. Bateson on that occasion (May 10) did not produce a single one of the many objections which are contained in his printed letter of June 2. The general impression which I gained at the meeting was that he could not think of any further objection to raise. The "vague diagrams" which he complained of appear not to be derived from my original paper (1909, *Archiv für Ent. Mech.*, vol. 28, Plate 16, Figs. 25 and 26), but—if I am not mistaken—from Plate ("Selektionsprinzip," 4th edition, p. 469, Fig. 95) in which my simple figures have been rather strongly altered and exaggerated.

Mr. Bateson must, therefore, have discovered by subsequent reflection all that during my demonstration and lecture he *did not* see. Otherwise it might have been possible for me to make him to see what he did not wish to see; I would certainly, for his benefit, have removed the Alytes specimen from the jar and he would have been able to view it—without obscuration by glass or background—from all sides under the lens: I treated it in this way during my stay in England for many colleagues (as, for example, for Mr. E. C. Boulenger and Sir Sidney Harmer).

On the occasion of the meeting of the Cambridge Natural History Society, I had at my disposal a Zeiss binocular microscope. Every unprejudiced observer could convince himself by its aid that the skin-area under discussion was of the nature of a nuptial pad—an area which according to Mr. Bateson was merely "a piece of thickened blackish-brown skin."

Numerous are the reasons which Mr. Bateson has given in order to be "absolved from basing broad conclusions on his testimony"; in a word, in order to deny

¹ NATURE, May 12, 1923, p. 639, column 2, line 10, should read "microscopic observation," not "microscopic."

the existence of the nuptial pads. First he questioned the existence of the pad; then he suggested it was merely a black patch of pigment; then, that it was present in only one specimen—consequently an accidental monstrosity; then he asserted that it was a shadow which appeared in the photograph; then, even that it had been produced by artificial retouching "in the wrong place," that is to say, on the outermost smallest finger, where in my untouched photograph (1919) some dirt had accidentally remained adhering. The microtome-sections of the pad-tissue, Mr. Bateson suggested, had been taken from another type of Anuran; then, since it appeared that homologous tissues of other species of Anura were of a different character, that these sections did not show genuine pad-tissue.

The most recent communication in which Mr. Bateson gives the impression which he received from the specimen which I demonstrated at the Linnean Society is capable of only two explanations, namely, either that Mr. Bateson is not an acute observer or that his theoretical views have affected his vision. In neither case can he escape the criticism that in describing "Dr. Kammerer's Alytes" he proceeded with a rashness unusual in a scientific man; especially when he makes slightly veiled accusations of "correcting Nature" against conscientious observers. We may now enumerate the points on which Mr. Bateson's "doubtful memory" has led him astray, namely:

(1) It is incorrect to say that my preparation of Alytes prevented a view of the dorsal aspect of the hand and only showed the palmar aspect—to make such a preparation it would have been necessary to fasten each finger flat against the substratum.

(2) It is incorrect to say that the black colour is restricted to the palmar aspect. (Why should Mr. Bateson assert this when he had not seen the dorsal aspect?) Actually the pads extend to the dorsal aspect and are therefore not "in the wrong place." It is curious to find Mr. Bateson prescribing to Nature the "right" place—Nature which has produced much more "astounding" and "curious creatures" (cf. *Trichobatrachus*) than my modest cultures of Alytes.

(3) It is incorrect to say "The right hand showed nothing special." On the inner side of the wrist-joint, on the insertion of the ball of the "thumb," there has been regenerated² a distinct dark pad—of course, not so large as that on the left hand.

(4) It is incorrect to say that the pad presents only "a dark uniform surface but no papillary or thorny structures." I send herewith an enlarged photograph in which "rugosities" can be seen on the edge of the pad with the naked eye. [I have verified this, but doubt very much whether the rugosities would appear in a print reproduced in NATURE.—E. W. M.] Unfortunately this photograph is taken from the palmar aspect; it was not foreseen that Mr. Bateson would criticise this, the most advantageous position, in order to deny the presence of the pad on the dorsal surface, and to call in question the pad-nature of the whole structure. It is probable that the majority of my English colleagues have no idea how difficult it is to obtain a satisfactory photograph in our impoverished Austria. Of course, at the very first opportunity I shall have the upper side photographed; perhaps Mr. Bateson in his desire for truth will provide the necessary camera and photographic materials.

Dozens of scientific men have seen the pads and are now convinced; only Mr. Bateson has seen nothing. Unfamiliar as he is with this special department, he expects to see the same as can be seen in

² Dr. Kammerer stated at the meeting of the Linnean Society that the original pad on the right hand had been removed for the purpose of making microtome sections.—E. W. M.

Rana agilis. His assertion that the pads of *Alytes obstetricans* are not pads because they have a different appearance from those of *Rana agilis* is as unreasonable as it would be to maintain that *Alytes obstetricans* is not a Batrachian because it does not look like *Rana agilis*.

Lastly, a few words on the question of adaptation. In my lecture I avoided speaking of adaptation because this term involves a hypothetical and teleological element: I feared that to use it might lead to endless unfruitful discussion. Unfortunately, I was unable to prevent this; Mr. Cunningham discussed his own theory of adaptation in a way that had little to do with the facts which I had cited. I definitely declined to enter into this subject in my reply, simply because it is not usual for the discussion to wander so far from the subject of the lecture. So far as the nuptial pads are concerned, may I refresh Mr. Bateson's memory so far as to remind him that not only my *Alytes* but also other Batrachians, and especially the Discoglossidæ (to which *Alytes* belongs), have pads on places which never come into contact with the female? *Bombinator pachypus*, for example, develops pads on two or three toes of the hind foot (cf. Schreiber, "Herpetologia Europæa," 1912, p. 175). Are these "in the wrong place" or "retouched" by Nature?

I willingly admit that the traditional explanation of the pads, namely, that they are produced by friction with the skin of the female, may possibly be a fable: for that reason I have referred to this view with reserve and scepticism in my paper (1919, pp. 331, 339, 353). It is true that the spread of the thickening to regions of the skin which in the copulatory act do not undergo friction, is no valid ground for rejecting the theory. Mr. Bateson has doubtless himself observed that pressure thickenings and blisters often extend beyond the original zone of irritation. But it is by no means impossible, although of course not proved (Kammerer, 1919, p. 340), that life in water produces the pads; if this were so we should have a case of direct passive production but not of active adaptation. The correctness of my observations and their relevance to the theory of heredity, is not affected whichever of the explanations is adopted.

PAUL KAMMERER.

DR. BATESON, in a letter to NATURE of June 2, raises the very interesting point as to whether the appearances alleged to be "nuptial pads" in *Alytes obstetricans* are really such. Whatever their nature, they are undoubtedly organised structures; and if they should prove not to be "nuptial pads," they will have to be regarded as a new and arbitrary feature which has appeared after subjection to an experimentally altered environment for two or three generations, and which persists for at least a few generations after a return to normal conditions. In other words, it would seem that Dr. Kammerer has had success in an experiment which is almost analogous to those ancient researches in which was attempted the reproduction by hereditary means of a surgically impressed modification.

However, Dr. Kammerer has clearly stated that in his opinion the only feature of the experiment which in any way justifies such a view is that the excrescences in question are not dependent for their development on the presence of a testis, and in this differ from the nuptial pads of the better known Amphibia Anura.

Dr. Bateson points to two details which make "the appearance quite unlike that of any natural *Brunftschwien*": first, that in *Alytes* there is a "dark uniform surface . . . without the dotting or stippling so obvious in true *Brunftschwien*";

secondly, that their position does not correspond to that of the nuptial pads in *Rana agilis*.

Lataste's excellent drawings (*Ann. Sci. Nat.* (6), tom. 3, pl. 11, 1876) show that a uniform blackness of the outer layer of the pad is a characteristic feature of the Discoglossidæ (to which *Alytes* belongs) and distinguishes them from other Anura. The fully developed pads of *Bufo vulgaris* are also uniformly black, and I have recently found that when such full hypertrophy of the outer epithelium is inhibited, as occasionally happens from obscure causes, it may be induced by making the male maintain a sexual embrace for a week or two. The same effect may be produced in the summer condition of the pad, and I have found that the hypertrophy takes place even when the male maintains his tonic embrace on thin air.¹

The pad of the *Alytes* "water-breed" also resembles that of the Discoglossid *Bombinator* in having a complete layer of black pigment in the cutis vera which would further contribute to the uniform dark appearance which *Alytes* so well and characteristically shows. Photographs show another interesting point. Very distinct connective-tissue papillæ are developed from the cutis vera in association with the epidermal spines. Such papillæ are but very slightly developed in the Discoglossidæ, though Lataste's picture of *Discoglossus* shows traces, whilst they are a characteristic feature of the pads of many other Batrachians.

The epidermal spines are very obvious in the intact specimen, as I have repeatedly seen both with lens and binocular microscope, and as many others have witnessed in my presence. Of course, they are practically impossible to photograph on account of the glistening of a wet specimen, but a photograph at least makes clear what areas of skin are affected. These include nearly the whole of the palm, the radial surface of the inner metacarpal and part of the first phalangeal joint of the thumb, and more or less of the ventral and radial surfaces of the forearm, passing over the dorso-radial margin of the inner carpal tubercle. The Discoglossidæ are remarkable for the very various positions in which the histological features of *Brunftschwien* may manifest themselves, on the chin, belly, thighs, toes of the feet even; in other words, they are not necessarily dependent on contact with the female for their development. Dr. H. Gadow has shown me his sketch of the nuptial pad in *Alytes cisternasii*, Bosca., where it is developed on the tip of the thumb, extending on the palmar surface. Even in the common toad I have frequently observed the nuptial rugosity extending on to the palmar surface of the inner carpal tubercle.

Questionable as it is to draw conclusions on anatomical points by analogy from other animals, it is even more unsafe to do so as regards their habits and postures; *Alytes* does not belong even to the same suborder as *Rana agilis*. De l'Isle (*Ann. Sci. Nat.* (6), tom. 3, p. 18), in his account of the cervical clasp of *Alytes*, says with regard to "les paumes," "les applique contre le cou de la femelle." Moreover, although he gives no definite description of the attitude of the hands during the inguinal clasp, he describes how, with the fingers interlaced, the two backwardly directed internal digits participate in the well-known chafing of the cloaca, which seems to me anatomically impossible if the hands are so much everted that the palms do not come in contact with the pubic region, the groins, or at least the thighs of the female.

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June 16.

¹ The surgical details of this experiment are of no importance in the present connexion.